

project proposal

1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
```

```

aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# read in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

2. Cleaning

sites object

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

Cleaning taxon_list

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

Cleaning water_quality

```

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names

```

```

clean_names() |>
# filtering join: only include sites in ncos_sites
semi_join(ncos_sites, by = "site") |>
# create new column to join with later data frames
unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(
  med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
  med_sal = median(salinity_ppt, na.rm = TRUE),
  med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

```

Cleaning aq_ins

```

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(
  date, sample_type, site,
  ostracod,
  copepod,
  hemiptera_corixidae_boatman,
  cladocera,
  nematode,
  diptera_ceratopogonidae,
  annelida_oligochaete,
  diptera_chironomid,
  annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format

```

```

pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

Focus Taxa

```

# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                 "Hemiptera Corixidae Boatman", "Nematode")

```

Figure 1: Invertebrate abundance as a function of DO

```
# create object sal_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)

# base layer: ggplot
DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
  # uses sal_taxa data frame
  mapping = aes(x = med_do, # x-axis: med_do
    y = log_ave_lit, # y-axis: log_ave_lit
    color = taxon)) + # color by taxon

# first layer: points show individual abundance measurements
geom_point(shape = 16, # shape is circle
  size = 2, # size is 2
  alpha = 0.7) + # opacity is 0.7

# separate panels for each taxon with free x- and y-axis scales
facet_wrap(~ taxon, scales = "free") +
# use non-default colors for points
scale_color_viridis_d() +
labs(x = "Median DO (mg/L)", # relabel x-axis
  y = "log + 1 transformed average abundance/L", # relabel y-axis
  # creates descriptive title
  title = "Invertebrate abundance as a function of DO") +
# uses non-default theme
theme_minimal() +
# remove redundant legend
theme(legend.position = "none")

# shows plot
```

DO_taxa_plot

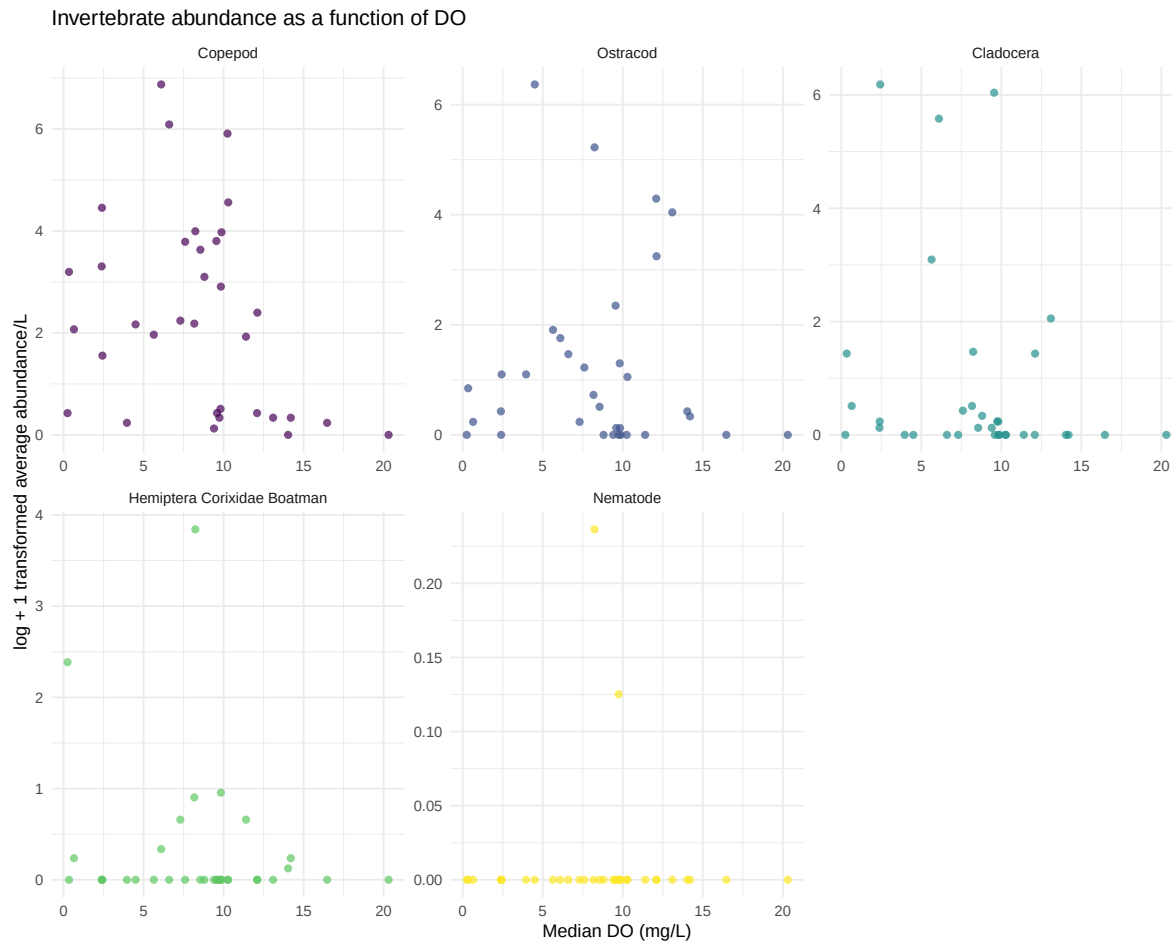


Figure 2: Variance of DO across sights

```
# base layer
DO <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_do)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
```

```

        width = 0.1,
        shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
             color = "red",
             size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median DO (mh/L)",
     title = "Site DO levels") +
# different theme
theme_bw()
# show plot
DO

```

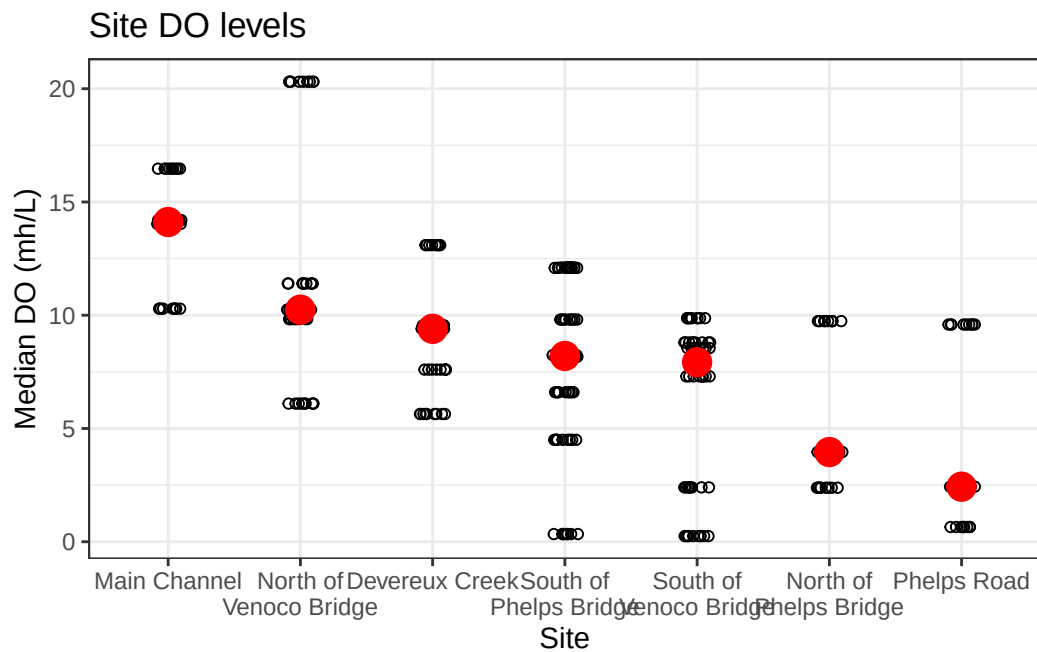


Figure 3: Aquatic Invert. Abundance Across Sites

```

# creating new object from clean aquatic inverts
abundance_by_site <- aq_ins_clean |>
  # group by taxon, site
  group_by(taxon, site) |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# base layer

abundance <- ggplot(data = abundance_by_site,
  # x-axis: site
  # y-axis: mean abundance/L
  # color by taxon
  mapping = aes(x = site_full,
    y = log_ave_lit,
    color = taxon)) +
  # first layer: points to represent observations
  geom_point() +
  # second layer: points to represent medians
  #stat_summary(geom = "point",
    #fun = "median",
    #size = 4,
    #color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
    y = "log + 1 transformed average abundance/L",
    title = "Aquatic Invert. Abundance Across Sites") +
  # different theme
  theme_bw()

abundance

```

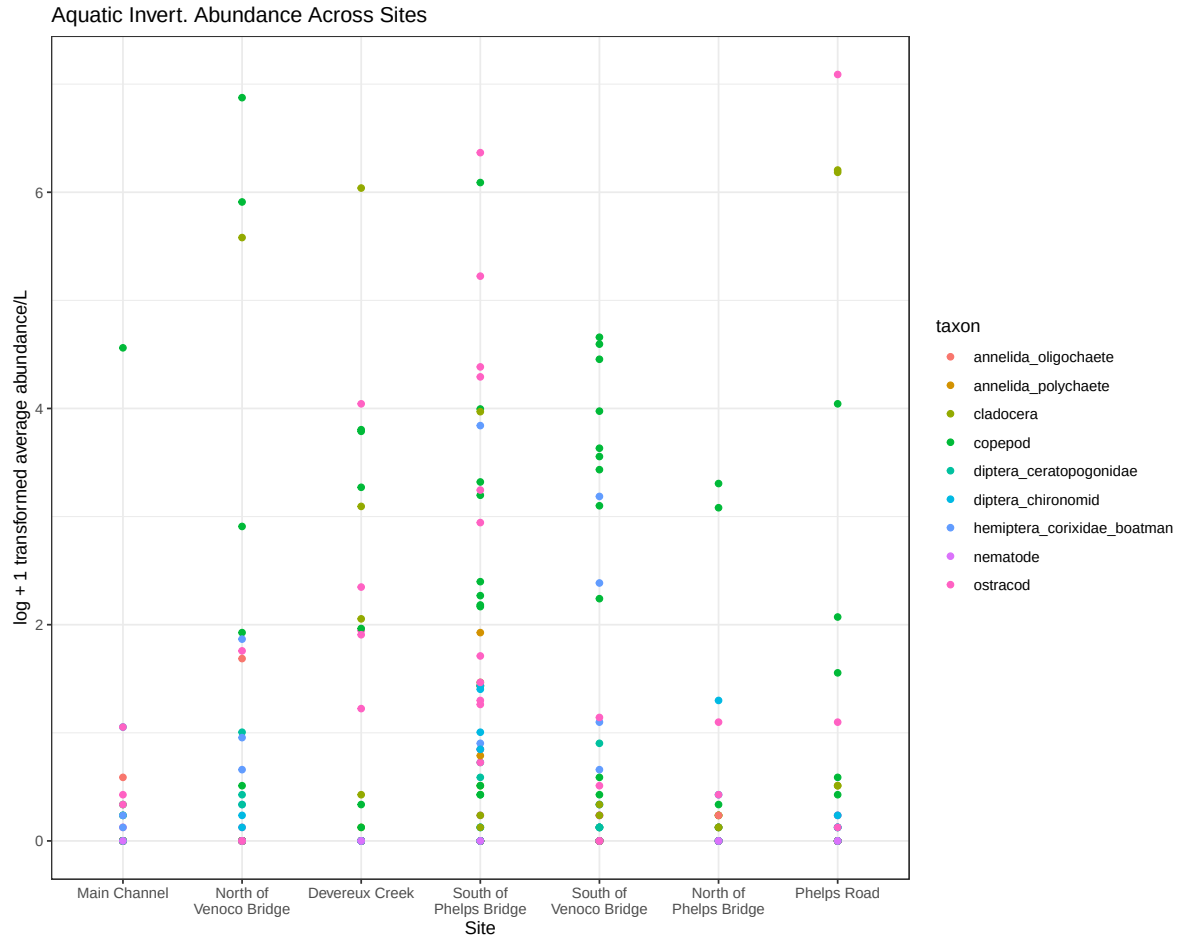



Figure 4: Abundance over time

```
# create object for plotting total abundance over time
abundance_time <- aq_ins_clean |>
  # group by sampling date and taxon
  group_by(date, taxon) |>
  # total abundance across all sites for each date and taxon
  summarize(total_ave_lit = sum(ave_lit, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create log-transformed total abundance column
  mutate(log_total_ave_lit = log(total_ave_lit + 1)) |>
  # make taxon names easier to read
```

```

mutate(taxon = str_replace_all(taxon, "_", " "),
       taxon = str_to_title(taxon)) |>
# order taxa by maximum total abundance
mutate(taxon = fct_reorder(taxon, total_ave_lit, .fun = max, .desc = TRUE)) |>
# include main taxon groups of focus
filter(taxon %in% c("Ostracod", "Copepod", 'Cladocera',
                  'Hemiptera Corixidae Boatman', 'Nematode'))

# base layer
abundance_time_plot <- ggplot(data = abundance_time,
                             # x-axis: sampling date
                             # y-axis: total log-transformed abundance
                             mapping = aes(x = date,
                                           y = log_total_ave_lit,
                                           color = taxon)) +

# points show total abundance on each sampling date
geom_point(size = 2,
           alpha = 0.7) +
# lines connect total abundance through time for each taxon
geom_line(linewidth = 0.8,
          alpha = 0.7) +
# separate panels for each taxon
facet_wrap(~ taxon, scales = "free_y") +
# use non-default colors
scale_color_viridis_d() +
# relabel axes and add title
labs(x = "Date",
     y = "log + 1 transformed total average abundance/L across sites",
     title = "Total invertebrate abundance over time") +
# use clean theme
theme_bw() +
# remove redundant legend
theme(legend.position = "none",
      axis.text.x = element_text(angle = 45,
                                  hjust = 1))

# show plot
abundance_time_plot

```

Total invertebrate abundance over time

